

HW1

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Q1

Let X , W , and Z denote matrices.

(a)

If $\mathcal{C}(X) = \mathcal{C}(W)$, show that $P_X = P_W$, i.e., the projection matrices must be the same.

Hint: Consider $(P_W - P_X)'(P_W - P_X)$.

Answer.

By Corollary 5 (4), of Lecture Notes part 1, $P_X X = X$. Also note: for any $v \in \mathcal{C}(X)$, for $v = Xa$, it follows that $P_X v = P_X Xa = Xa = v$ (and similarly for P_W).

Now, to answer the question. If $\mathcal{C}(X) = \mathcal{C}(W)$, then every column of W lies in $\mathcal{C}(X)$ (and vice versa). Consequently, $P_X W = W$ and $P_W X = X$.

It then follows

$$P_W P_X = P_W X (X' X)^{-} X' = X (X' X)^{-} X' = P_X,$$

and

$$P_X P_W = P_X W (W' W)^{-} W' = W (W' W)^{-} W' = P_W.$$

Using symmetry and idempotency of projection matrices P_X and P_W ,

$$(P_W - P_X)'(P_W - P_X) = P_W - P_W P_X - P_X P_W + P_X = P_W - P_X - P_W + P_X = 0.$$

By Lemma 1.2, $A' A = 0 \Rightarrow A = 0$. It then follows that $P_X = P_W$. //

(b)

If X is an $n \times n$ matrix of rank n , show $P_X = I_n$ (i.e., the $n \times n$ identity matrix).

Answer.

If X is $n \times n$ of rank n , X^{-1} exists and is unique (nonsingular as X is full-rank).

By Lemma 2.3, $\text{rank}(X' X) = \text{rank}(X) = n$, so $X' X$ is nonsingular with unique generalized inverse, i.e. $(X' X)^{-} = (X' X)^{-1}$.

Then, noting the product rule for matrix inverses (apparently called *The Shoes and Socks Property*!?)

$$P_X = X (X' X)^{-} X' = X (X' X)^{-1} X' = X X^{-1} (X')^{-1} X' = I_n. //$$

(c)

Let $\mathcal{C}^\perp(X)$ denote the orthogonal complement of $\mathcal{C}(X)$ (i.e., $\underline{y} \in \mathcal{C}^\perp(X)$ if and only if $\underline{x}'\underline{y} = 0$ for any column $\underline{x}$ of X). If a vector $\underline{b} \in \mathcal{C}(X)$ and $\underline{b} \in \mathcal{C}^\perp(X)$, show that $\underline{b}$ must equal the zero vector (this also implies that a non-zero vector in $\mathcal{C}(X)$ must be linearly independent from a non-zero vector in $\mathcal{C}^\perp(X)$).

Hint: Write $\underline{b} = X\underline{a}$ and consider $\underline{b}'\underline{b}$.

Answer.

If $\underline{b} \in \mathcal{C}(X)$, it follows that $\underline{b} = X\underline{a}$ for some $\underline{a}$.

If we also presuppose that $\underline{b} \in \mathcal{C}^\perp(X)$, it follows that $X'\underline{b} = \underline{0}$.

It then follows that $\underline{b}'\underline{b} = (X\underline{a})'\underline{b} = \underline{a}'X'\underline{b} = 0$.

However, since $\underline{b}'\underline{b} = \sum_i b_i^2 = 0$, it follows that $b_i = 0$, for all i , and we must conclude that $\underline{b} = \underline{0}$. //

(d)

Let X be an $n \times p$ matrix of rank $r \geq 1$, and Z be an $n \times (n - r)$ matrix with columns formed by $n - r$ linearly independent vectors in $\mathcal{C}^\perp(X)$. Show that $P_Z = I_n - P_X$.

Hint: Put r linearly independent columns of X along with the columns of Z into one $n \times n$ matrix $\tilde{X}$ (if $n = r$, technically Z is just a zero vector). Compute $P_{\tilde{X}}$ and use (a)-(b).

Answer.

Let X_1 ($n \times r$) consist of r linearly independent columns of X , such that $\mathcal{C}(X_1) = \mathcal{C}(X)$ and $P_{X_1} = P_X$ (see part (a) for the latter result). We split the proof into two cases: (1) the degenerate case ($r = n$) and (2) the general case, where we embed X_1, Z into a nonsingular $\tilde{X} = (X_1 \ Z)$ and compute $P_{\tilde{X}}$.

Degenerate case ($r = n$): Since $r = n$, $\dim \mathcal{C}^\perp(X) = n - r = 0$, so $\mathcal{C}^\perp(X) = \{\underline{0}\}$ ($\{\underline{0}\}$ being the zero vector) and Z has zero columns, giving $\tilde{X} = X_1$. Since X_1 is $n \times n$ of rank n (nonsingular), applying part (b) gives $P_X = P_{X_1} = I_n$, and $P_Z = I_n - P_X = 0$ holds.

General case: Let $\tilde{X} = (X_1 \ Z)$ and consider $1 \leq r < n$.

Nonsingularity: X_1 's columns are independent and span $\mathcal{C}(X)$ ($\dim r$); Z 's columns are independent and span $\mathcal{C}^\perp(X)$ ($\dim n - r$). By (c), $\mathcal{C}(X) \cap \mathcal{C}^\perp(X) = \{\underline{0}\}$, so any linear dependence among the combined columns of $\tilde{X}$ would force a nonzero vector into $\mathcal{C}(X) \cap \mathcal{C}^\perp(X)$ and arrive at a contradiction. It then follows that all n columns of $\tilde{X}$ are independent, i.e., $\text{rank}(\tilde{X}) = n$.

Block structure: Columns of Z lie in $\mathcal{C}^\perp(X)$, so $X_1'Z = 0 = Z'X_1$, giving

$$\tilde{X}'\tilde{X} = \begin{pmatrix} X_1'X_1 & 0 \\ 0 & Z'Z \end{pmatrix}, \quad (\tilde{X}'\tilde{X})^{-1} = \begin{pmatrix} (X_1'X_1)^{-1} & 0 \\ 0 & (Z'Z)^{-1} \end{pmatrix},$$

where both diagonal blocks are nonsingular by Lemma 2.3 (noted previously as $\text{rank}(X'X) = \text{rank}(X)$).

Hence

$$P_{\tilde{X}} = \tilde{X}(\tilde{X}'\tilde{X})^{-1}\tilde{X}' = X_1(X_1'X_1)^{-1}X_1' + Z(Z'Z)^{-1}Z' = P_{X_1} + P_Z = P_X + P_Z.$$

Since $\tilde{X}$ is nonsingular, part (b) gives $P_{\tilde{X}} = I_n$, so $P_Z = I_n - P_X$. //

(e)

For a vector $\underline{y}$, show that $P_X \underline{y} = 0$ if and only if $\underline{y} \in \mathcal{C}^\perp(X)$.

Hint: use (d) and multiply $I = I - P_X + P_X$ by $\underline{y}$

Answer.

The general approach is to show both directions hold (*if* and *only if*). To that end:

($\Leftarrow$): Suppose $\underline{y} \in \mathcal{C}^\perp(X) = \mathcal{C}(Z)$.

Since $\underline{y} \in \mathcal{C}(Z)$, it follows that $P_Z \underline{y} = \underline{y}$.

By part (d), $I = P_X + P_Z$.

Multiplying by $\underline{y}$,

$$\underline{y} = I\underline{y} = P_X \underline{y} + P_Z \underline{y} = P_X \underline{y} + \underline{y} \implies P_X \underline{y} = \underline{0}.$$

($\Rightarrow$): If $P_X \underline{y} = \underline{0}$, then by part (d), $\underline{y} = I\underline{y} = P_X \underline{y} + P_Z \underline{y} = P_Z \underline{y} \in \mathcal{C}(Z) = \mathcal{C}^\perp(X)$. //

(f)

Show that $P_X + P_W$ is a projection matrix if and only if $X'W = 0$ (the columns of X and W must be orthogonal).

Hint: If $P_X + P_W$ is a projection matrix, one can show $P_X P_W = -P_W P_X$ must hold; this relation implies that any vector $\underline{y}$ that is in both $\mathcal{C}(X)$ and $\mathcal{C}(W)$ must be zero (why?). Further, for $M = P_X P_W$, the relation also gives $M' M = P_X P_W P_W$ and $M' M = P_W P_X P_W$, meaning any column of $M' M$ belongs to $\mathcal{C}^\perp(X)$ and to $\mathcal{C}^\perp(W)$.

Key Results (this result is used in parts (f) and (g), with proof provided here once for completeness)

Result 1: (converse of Corollary 5 from Lecture Notes 1) M is symmetric and idempotent $\iff M = P_A$ for some A .

Proof. We take the standard approach of proving both directions.

($\Rightarrow$): We can directly note Corollary 5.

($\Leftarrow$): If $M' = M$ and $M^2 = M$, then $M \cdot M \cdot M = M$, so M is a generalized inverse of $M' M = M$. Thus $P_M = M(M' M)^- M' = M^3 = M$. //

Result 2: Since $P_X + P_W$ is automatically symmetric (sum of symmetric matrices is symmetric), applying the above result gives

$$P_X + P_W \text{ is a projection matrix } \iff P_X + P_W \text{ is idempotent.}$$

Now, on to the question at hand.

Answer.

We prove $P_X + P_W$ is a projection matrix $\iff X'W = 0$ by establishing each direction separately.

($\Leftarrow$): Suppose $X'W = 0$, so also $W'X = (X'W)' = 0$. Then

$$P_X P_W = X(X'X)^- X'W(W'W)^- W' = X(X'X)^- \underbrace{(X'W)}_{=0} (W'W)^- W' = 0,$$

and symmetrically $P_W P_X = 0$.

Hence $P_X + P_W$ is idempotent, as

$$(P_X + P_W)^2 = P_X^2 + P_X P_W + P_W P_X + P_W^2 = P_X + 0 + 0 + P_W = P_X + P_W,$$

By Result 2 noted above, $P_X + P_W$ is a projection matrix.

($\Rightarrow$): Suppose $P_X + P_W$ is a projection matrix. By Result 2, $P_X + P_W$ is idempotent, so

$$P_X + P_X P_W + P_W P_X + P_W = P_X + P_W \implies P_X P_W = -P_W P_X. \quad (1)$$

The rest of the proof (in this direction) is more involved as we use (1) to show $M := P_X P_W$ satisfies $M' M = 0$, leading us to conclude $M = 0$.

Goal: $\mathcal{C}(X) \cap \mathcal{C}(W) = \{\underline{0}\}$.

Let $\underline{y} \in \mathcal{C}(X) \cap \mathcal{C}(W)$. It follows that $P_X \underline{y} = \underline{y}$ (since $\underline{y} \in \mathcal{C}(X)$) and $P_W \underline{y} = \underline{y}$ (since $\underline{y} \in \mathcal{C}(W)$). Applying (1) to $\underline{y}$:

$$\underline{y} = P_X(P_W \underline{y}) = (P_X P_W) \underline{y} = -(P_W P_X) \underline{y} = -P_W(P_X \underline{y}) = -\underline{y},$$

forcing $\underline{y} = \underline{0}$.

Goal: $\mathcal{C}(M' M) \subseteq \mathcal{C}(X) \cap \mathcal{C}(W)$.

Since P_X, P_W are symmetric (see Corollary 5), $M' = (P_X P_W)' = P_W' P_X' = P_W P_X$, so

$$M' M = P_W P_X P_X P_W = P_W (P_X^2) P_W = P_W P_X P_W,$$

using idempotency of P_X .

We compute this product two ways:

- (i) Grouping as $P_W(P_X P_W) = P_W M$: since the columns of a product AB are linear combinations of the columns of A , $\mathcal{C}(M' M) = \mathcal{C}(P_W M) \subseteq \mathcal{C}(P_W) = \mathcal{C}(W)$.
- (ii) Grouping as $(P_W P_X) P_W$ and applying $(*)$ ($P_W P_X = -P_X P_W$):

$$M' M = (P_W P_X) P_W \stackrel{(*)}{=} -P_X (P_W P_W) = -P_X P_W = -M,$$

using idempotency of P_W . Hence $\mathcal{C}(M' M) = \mathcal{C}(-M) = \mathcal{C}(M) = \mathcal{C}(P_X P_W) \subseteq \mathcal{C}(P_X) = \mathcal{C}(X)$.

Combining (i) and (ii), every column of $M' M$ lies in $\mathcal{C}(X) \cap \mathcal{C}(W)$, which is $\{\underline{0}\}$.

Hence $M' M = 0$.

Goal: $M = 0$.

By Lemma 1.2 ($M' M = 0 \implies M = 0$ for any real matrix M — the same column-by-column argument as in part (c)), $M' M = 0$ gives $M = P_X P_W = 0$. Then (1) gives $P_W P_X = -P_X P_W = 0$ as well.

Goal: $X' W = 0$.

Right-multiplying $P_X P_W = 0$ by W and using the property $P_W W = W$:

$$0 = (P_X P_W) W = P_X (P_W W) = P_X W.$$

Left-multiplying X' and using $X' P_X = X'$ (noting the transpose of the projection matrix is the projection matrix):

$$0 = X'(P_X W) = (X' P_X) W = X' W.$$

Hence $X' W = 0$, completing the proof of $(\Rightarrow)$. //

(g)

Show that $P_X - P_W$ is a projection matrix if and only if $\mathcal{C}(W) \subset \mathcal{C}(X)$.

Hint: If $P_X - P_W$ is a projection matrix then $(I - P_X)P_W = -P_W(I - P_X)$ must hold; then the proof is kind of the same as in (f), using that (from (d)) $P_Z = I_n - P_X$ holds, where Z is a matrix with linearly independent columns that span $\mathcal{C}^\perp(X)$.

Answer.

$(\Leftarrow)$: If $\mathcal{C}(W) \subseteq \mathcal{C}(X)$, then since Z 's columns span $\mathcal{C}^\perp(X)$, they're orthogonal to $\mathcal{C}(X) \supseteq \mathcal{C}(W)$, so $Z' W = 0$. By part (f), $P_Z P_W = P_W P_Z = 0$ and $P_Z + P_W$ is symmetric and idempotent.

Then $P_X - P_W = (I - P_Z) - P_W = I - (P_Z + P_W)$, and for a symmetric and idempotent Q , $I - Q$ is also symmetric and idempotent, so $P_X - P_W$ is a projection matrix (applying Result 2 from the preface to part (f)).

$(\Rightarrow)$: If $P_X - P_W$ is a projection matrix, idempotency gives

$$P_X - P_X P_W - P_W P_X + P_W = P_X - P_W \implies P_X P_W + P_W P_X = 2P_W.$$

Substituting $P_X = I - P_Z$:

$$(I - P_Z)P_W + P_W(I - P_Z) = 2P_W \implies P_Z P_W = -P_W P_Z,$$

which is exactly relation (1) from part (f) with Z in place of X .

The identical computation as in part (f) and gives $Z' W = 0$, i.e. $\mathcal{C}(W) \subseteq \mathcal{C}^\perp(Z)$.

Since $\mathcal{C}(X) \subseteq \mathcal{C}^\perp(Z)$ by construction and $\dim \mathcal{C}^\perp(Z) = n - (n - r) = r = \dim \mathcal{C}(X)$, equal dimension plus containment gives $\mathcal{C}^\perp(Z) = \mathcal{C}(X)$. Hence $\mathcal{C}(W) \subseteq \mathcal{C}(X)$, proving $(\Rightarrow)$. //

2.

Suppose that $\mathcal{C}(C)$ is a subspace of $\mathcal{C}(A)$ and that $\mathcal{R}(D)$ is a subspace of $\mathcal{R}(A)$.

(a)

Show that $AA^-C = C$ and $DA^-A = D$.

Answer.

Since $\mathcal{C}(C) \subseteq \mathcal{C}(A)$, Lemma 1.6 (i) gives $C = AF$ for some matrix F .

Using the definition of a generalized inverse, i.e., $AA^-A = A$, it then follows

$$AA^-C = AA^-(AF) = (AA^-A)F = AF = C.$$

Also, since $\mathcal{R}(D) \subseteq \mathcal{R}(A)$, Lemma 1.6 (ii) gives $D = GA$ for some matrix G .

Similarly, using the definition of a generalized inverse, it then follows

$$DA^-A = (GA)A^-A = G(AA^-A) = GA = D. //$$

(b)

Using (a), show that

$$\begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix} \begin{bmatrix} A & C \\ D & B \end{bmatrix} \begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & Q \end{bmatrix},$$

where $Q = B - DA^-C$, and I denotes the identity matrix of appropriate dimensions.

Answer.

As given, we must compute $\begin{bmatrix} A & C \\ D & B \end{bmatrix} \begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix}$ blockwise. The following refers to the resultant matrix (we start with multiplication of the second and third matrices on the left-hand side):

- top-left: $AI + C \cdot 0 = A$
- top-right: $A(-A^-C) + CI = -AA^-C + C = -C + C = 0$ (by using part (a))
- bottom-left: $DI + B \cdot 0 = D$
- bottom-right: $D(-A^-C) + BI = B - DA^-C = Q$

Taken together, this leaves us:

$$\begin{bmatrix} A & C \\ D & B \end{bmatrix} \begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ D & Q \end{bmatrix}.$$

Now left-multiply by $\begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix}$:

- top-left: $IA + 0 \cdot D = A$
- top-right: $I \cdot 0 + 0 \cdot Q = 0$
- bottom-left: $-DA^-A + ID = -D + D = 0$ (again, using the results of part (a))
- bottom-right: $-DA^- \cdot 0 + IQ = Q$

Assembling the resultant matrix, we have:

$$\begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix} \begin{bmatrix} A & 0 \\ D & Q \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & Q \end{bmatrix} //$$

(c)

Using (b) with the fact that the two matrices defined by “ I on the diagonals” are square and invertible in (b), show that

$$\begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix} \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} \begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix}$$

is a generalized inverse of

$$M \equiv \begin{bmatrix} A & C \\ D & B \end{bmatrix}.$$

Answer.

Let $L = \begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix}$ (the left-most matrix of the left-hand side equation), $R = \begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix}$ (the right-most matrix of the left-hand side equation), and $\tilde{M} = \begin{bmatrix} A & 0 \\ 0 & Q \end{bmatrix}$.

We confirm by direct computation. First, we note that L, R are invertible, with

$$L^{-1} = \begin{bmatrix} I & 0 \\ DA^- & I \end{bmatrix}, \quad R^{-1} = \begin{bmatrix} I & A^-C \\ 0 & I \end{bmatrix},$$

since

$$LL^{-1} = \begin{bmatrix} I & 0 \\ -DA^- + DA^- & I \end{bmatrix} = I, \quad RR^{-1} = \begin{bmatrix} I & -A^-C + A^-C \\ 0 & I \end{bmatrix} = I.$$

By part (b), $LMR = \tilde{M}$, so $M = L^{-1}\tilde{M}R^{-1}$ by multiplication of L^{-1} and R^{-1} on the left-side and right-side respectively.

Let $G = R \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} L$.

Then, since $R^{-1}R = I$ and $LL^{-1} = I$,

$$MGM = (L^{-1}\tilde{M}R^{-1}) \left(R \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} L \right) (L^{-1}\tilde{M}R^{-1}) = L^{-1}\tilde{M}(R^{-1}R) \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} (LL^{-1})\tilde{M}R^{-1} = L^{-1}\tilde{M} \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} \tilde{M}R^{-1}$$

Consequently, using $AA^-A = A$ and $QQ^-Q = Q$, we have

$$\tilde{M} \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} \tilde{M} = \begin{bmatrix} AA^-A & 0 \\ 0 & QQ^-Q \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & Q \end{bmatrix} = \tilde{M},$$

Hence $MGM = L^{-1}\tilde{M}R^{-1} = M$, and we conclude that G is a generalized inverse of M . //

(d)

Show that the above generalized inverse matches that given in (i) of Theorem 1.4.

Answer.

From part (c), we have $G = R \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} L$. We show this matches the form given in Theorem 1.4 (i) by computation.

First,

$$\begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} L = \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} \begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix} = \begin{bmatrix} A^- & 0 \\ -Q^-DA^- & Q^- \end{bmatrix}.$$

Left-multiplying by R then gives us an expression of G :

$$G = R \begin{bmatrix} A^- & 0 \\ -Q^-DA^- & Q^- \end{bmatrix} = \begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix} \begin{bmatrix} A^- & 0 \\ -Q^-DA^- & Q^- \end{bmatrix} = \begin{bmatrix} A^- + A^-CQ^-DA^- & -A^-CQ^- \\ -Q^-DA^- & Q^- \end{bmatrix},$$

Some notes on the above computation, by block:

- top-left = $A^- + (-A^-C)(-Q^-DA^-) = A^- + A^-CQ^-DA^-$
- top-right = $(-A^-C)Q^- = -A^-CQ^-$
- bottom-left = $-Q^-DA^-$
- bottom-right = Q^-

Confirming that $G = R \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} L$ matches the form of the matrix given in Theorem 1.4 (i). //

(e)

Show that the conditions imposed on the matrices B , C , and D in part (ii) of Theorem 1.4 are satisfied if there exists a matrix $Z = (Z_1, Z_2)$ for which

$$M = Z'Z = (Z_1, Z_2)'(Z_1, Z_2) = \begin{bmatrix} Z_1'Z_1 & Z_1'Z_2 \\ Z_2'Z_1 & Z_2'Z_2 \end{bmatrix} = \begin{bmatrix} A & C \\ D & B \end{bmatrix}.$$

Answer.

By Lemma 1.4 (using $X = Z_2$), we have $Z_2'Z_2(Z_2'Z_2)^-Z_2' = Z_2'$ (i.e., $Z_2(Z_2'Z_2)^-$ is a generalized inverse of Z_2'), such that $Z_2' = (Z_2'Z_2)[(Z_2'Z_2)^-Z_2']$.

By Lemma 1.6 (i), using $(\Leftarrow)$, we have $\mathcal{C}(Z_2') \subseteq \mathcal{C}(Z_2'Z_2)$.

Conversely, as $Z_2'Z_2 = Z_2' \cdot Z_2$, we know $\mathcal{C}(Z_2'Z_2) \subseteq \mathcal{C}(Z_2')$ (using Lemma 1.6 (i) again, in the same $(\Leftarrow)$ direction as above, using $A = Z_2'$, $C = Z_2$).

Hence

$$\mathcal{C}(Z_2') = \mathcal{C}(Z_2'Z_2) = \mathcal{C}(B).$$

Since $D = Z_2'Z_1$, each column of D is Z_2' times a column of Z_1 , so by Lemma 1.6 (i), $\mathcal{C}(D) \subseteq \mathcal{C}(Z_2') = \mathcal{C}(B)$, matching the first condition of Theorem 1.4 (ii).

For the second condition, note $C = Z_1'Z_2$, such that $C' = (Z_1'Z_2)' = Z_2'Z_1 = D$ and $B' = (Z_2'Z_2)' = Z_2'Z_2 = B$.

Using $\mathcal{R}(X) = \mathcal{C}(X')$ (the row space of X is the column space of X'),

$$\mathcal{R}(C) = \mathcal{C}(C') = \mathcal{C}(D) \subseteq \mathcal{C}(B) = \mathcal{C}(B') = \mathcal{R}(B),$$

matching the second and final condition of Theorem 1.4 (ii). //

3.

For each of the following statements, indicate whether the statement is true or false, and explain.

(a)

If $Ax = y$ has a unique solution, then A must be a nonsingular matrix.

False.

The term *nonsingular* is defined only for square matrices, so the statement as given presupposes something that is not guaranteed. Uniqueness of a solution to $A\mathbf{x} = \mathbf{y}$ depends only on whether A has full column rank, and that condition is meaningful for matrices of any dimensions.

To that end, as a counterexample let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

A is 3×2 with full column rank, so for *any* $\mathbf{y}$ in the column space of A , the system $A\mathbf{x} = \mathbf{y}$ has the unique solution. Yet A is not square, so it is neither singular nor nonsingular. As such, the statement above is false not because A happens to be singular, but because the term doesn't apply to it. //

Note: If we restricted the matrix A to be a square matrix, the statement is true. In that scenario the matrix is full rank, is invertible, and so uniqueness forces nonsingularity.

(b)

If G is a generalized inverse of A , then $GA = I$, where I denotes the identity matrix of appropriate dimensions.

False.

The defining property of a generalized inverse is $AGA = A$; however, this does not force $GA = I$. As a counterexample, let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

A is symmetric, singular ($\det A = 0$), and idempotent ($A^2 = A$). Since $AGA = AIA = A^2 = A$, and $G = I$ is a generalized inverse of A .

However,

$$GA = IA = A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

So the statement is shown to be false.//

(c)

If A has a generalized inverse G such that $GA = I$, then A must be a (square) nonsingular matrix.

False.

The condition $GA = I_m$ forces A (with dimension $n \times m$) to have full column rank, but this is again a rank condition, not a requirement that $n = m$ (see part (a)). Thus, we proceed with a counterexample as proof.

Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

A is a 3×2 matrix of full column rank, such that $A^\top A$ is 2×2 and nonsingular. Setting $G = (A^\top A)^{-1} A^\top$ gives $GA = (A^\top A)^{-1} A^\top A = I_2$ and $AGA = A(A^\top A)^{-1} A^\top A = A$, so G is a generalized inverse of A satisfying $GA = I$.

Yet A itself is 3×2 (not square) and cannot be nonsingular (and it cannot be singular too).//

(d)

If A is a symmetric matrix, then all its generalized inverses are also symmetric.

False.

Since the statement is false, it suffices to show a counterexample for proof. To that end, we need only identify a single symmetric matrix A for which some generalized inverse G fails to be symmetric.

To that end, let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = A',$$

and take

$$G = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad G' = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}.$$

Direct computation gives

$$AGA = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = A,$$

so G is a valid generalized inverse of A , but $G \neq G^\top$. //

(e)

If A is a symmetric matrix, then it has at least one symmetric generalized inverse.

True.

Unlike the previous parts of this question, a counterexample alone is not sufficient. We must establish existence.

By Theorem 1.1, A has at least one generalized inverse; call it G , so $AGA = A$. Since A is symmetric, the Corollary to Lemma 1.1 gives that G' is also a generalized inverse of A , i.e., $AG'A = A$.

Define $H = \frac{1}{2}(G + G')$.

Then

$$AHA = \frac{1}{2}(AGA + AG'A) = \frac{1}{2}(A + A) = A,$$

so H is a generalized inverse of A .

Moreover, H is symmetric by construction, as

$$H' = \frac{1}{2}(G' + G'') = \frac{1}{2}(G' + G) = H.$$

Thus H is a symmetric generalized inverse of A , proving that every symmetric matrix has at least one symmetric generalized inverse. //

4.

For each of the following questions, give, if possible, a 3×3 **nondiagonal** matrix A that has the stated property.

(a)

A is a generalized inverse of A .

Answer

It is possible, and to illustrate take

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

Generally, A being a generalized inverse of itself means $A \cdot A \cdot A = A$, i.e. $A^3 = A$. This holds for any idempotent matrix, since $A^2 = A$ implies $A^3 = A^2 \cdot A = A \cdot A = A^2 = A$.

So we simply need to identify nondiagonal matrix that satisfies $A^2 = A$. The matrix stated above suffices, as:

$$AA = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = A. //$$

(b)

A' is a generalized inverse of A .

Answer

It is possible, and as an example take

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Generally, we need to identify a matrix A such that $AA^T A = A$. This condition holds when $A^T = A^{-1}$.

Since $A^T A = I$, we have $A^T = A^{-1}$, and so $AA^T A = AA^{-1}A = A$. Using the matrix given above:

$$A^T = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = A^{-1},$$

and

$$AA^T A = AA^{-1}A = A. //$$

(c)

The identity matrix I is a generalized inverse of A .

Answer

It is possible, and to illustrate take the matrix used in part (a):

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Generally, the necessary condition is that $AIA = A^2 = A$ holds, i.e., that A is idempotent. That is to say, the identity matrix I is a generalized inverse of A when the matrix A is idempotent.

But for the purpose of this example, we verify:

$$AIA = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = A. //$$

(d)

Every 3×3 matrix is a generalized inverse of A .

Answer

It is **not possible** (to find a matrix A).

Let A be an arbitrary nondiagonal 3×3 matrix, and take $G = 0$. Then

$$AGA = A \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

regardless of the entries of A . So $AGA = A$ only if $A = 0$. However, the zero matrix is diagonal, contradicting the assumption that A is nondiagonal. Hence $G = 0$ fails to be a generalized inverse for *every* nondiagonal A , and no such A exists.//

(e)

Every generalized inverse of A is nonsingular.

Answer

It is possible, and to illustrate take the (nondiagonal, also nonsingular) matrix:

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \det(A) = 1 \neq 0.$$

Its inverse is

$$A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and we verify

$$AA^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I.$$

Generally, since A is nonsingular, suppose G is *any* generalized inverse of A , i.e. $AGA = A$. Left-multiplying by A^{-1} gives $GA = I$; right-multiplying this by A^{-1} gives $G = A^{-1}$. So $G = A^{-1}$ is forced — it is the unique generalized inverse of A . Since $\det(A^{-1}) = \det(A)^{-1} = 1 \neq 0$, this unique generalized inverse is nonsingular, and hence every generalized inverse of A (there being only the one) is nonsingular.//

5.

For any matrix A , let B and C be any two matrices that satisfy $A'AB = A'$ and $AA'C' = A$ (the existence of such B and C follows from Corollary 2 to Lemma 1.4). Let $G = CAB$. Show that G satisfies the following four conditions:

(i)

$$AGA = A$$

Answer.

We invoke Corollary 1 to Lemma 1.5: for any matrix X and any matrix B satisfying $X'XB = X'$, it holds that $XBX = X$.

(1) Take $X = A$: given $A'AB = A'$, with $X = A$, it then follows that $X'XB = X'$.

The corollary gives $XBX = X$, i.e.

$$ABA = A. \tag{1}$$

(2) Take $X = A'$: given $AA'C' = A$, substitute B with C' :

$$X'XC' = AA'C' = AA'C' = A = X',$$

So $X'XC' = X'$ holds, and application of the corollary gives $XC'X = X$, i.e., $A'C'A' = A'$.

Now transpose both sides (using $(PQR)' = R'Q'P'$ with $P = A', Q = C', R = A'$):

$$(A'C'A')' = (A')'(C')'(A')' = A C A,$$

Using $(A')' = A$, we then have

$$ACA = A. \tag{2}$$

(The rest of the proof)

Now, with $G = CAB$,

$$AGA = A(CAB)A = (ACA)(BA) = A(BA) = ABA = A. //$$

(ii)

$$GAG = G$$

Answer.

Using (1) from part (i),

$$GA = (CAB)A = C(ABA) = CA, \tag{3}$$

and similarly using (2) from part (i),

$$AG = A(CAB) = (ACA)B = AB. \tag{4}$$

Taking everything together,

$$GAG = (GA)G = (CA)(CAB) = C(ACA)B = CAB = G. //$$

(iii)

$$(GA)' = GA$$

Answer.

Take $X = A'$, so $X' = A$.

As in part (i), C' satisfies $X'XC' = AA'C' = A = X'$. Note and then apply Corollary 4 to Lemma 1.5, which states:

Let X be any matrix and let B be any matrix that satisfies $X'XB = X'$. Then, the matrix XB is symmetric.

We thus note that $XC' = A'C'$ is symmetric, i.e.:

$$(A'C')' = A'C'.$$

Since $(A'C')' = (C')'(A')' = CA$, it follows that $CA = A'C'$.

Similarly, the standard transpose rule gives $(CA)' = A'C'$.

Combining these two statements,

$$(CA)' = A'C' = CA.$$

By (3) in part (ii), $GA = CA$, so $(GA)' = (CA)' = CA = GA$.//

(iv)

$$(AG)' = AG$$

Answer.

Take $X = A$.

Since B satisfies $X'XB = A'AB = A' = X'$, Corollary 4 to Lemma 1.5, as noted in part (iii), gives that $XB = AB$ is symmetric, i.e.:

$$(AB)' = AB.$$

By (4) in part (ii), $AG = AB$, so $(AG)' = (AB)' = AB = AG$.//